

How did Ivory Coast win AFCON 2023 despite sacking their manager during the Tournament, and was it deserved?

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1 My Resources

Live dashboard: <https://nahal12345.github.io/Sports-Data-and-Analytics-Assignment-/>

Repository: <https://github.com/Nahal12345/Sports-Data-and-Analytics-Assignment->

2 Introduction

AFCON 2023 was hosted in Côte d'Ivoire (Ivory Coast), and it saw the hosts lift the title. This victory was remarkable not just because it was a home tournament, but the team were on the brink of elimination during the group stage. After their win in the first game, Ivory Coast followed that by losing two consecutive games to Nigeria and Equatorial Guinea which saw them finish in 3rd place, narrowly qualifying due to them being the 4th best 3rd place finishers. Due to this failure in the group stage, the Ivorian Football Federation took the decision to sack the head coach Jean-Louis Gasset and replace him with Emerese Faé, who was the Ivory Coast under 23 coach.

Following this there was an unexpected revival, with Ivory Coast which led to AFCON victory. Faé's side beat Senegal in the Round of 16, Mali in the Quarter Final, Democratic Republic of Congo in the Semi Final, and then defeated Nigeria in the final, which was one of the team's that they had previously lost to in the group stage. They didn't only just beat them, they beat them convincingly as they dominated the final. The transformation between the two managers was visible, but the question that this report seeks to answer is whether the data supports the

outcome: How did Ivory Coast win AFCON 2023 despite sacking their manager during the Tournament, and was it deserved?

The question is interesting because football tournaments can be won due to pure luck and individual brilliance whilst also posting poor underlying numbers. On the contrast, a dominant team can lose to a single moment. Expected Goals (xG), defensive action metrics, attacking metrics and shot quality data offers a way to cut through the narrative and allows us to analyse what exactly is happening with the players on the pitch.

The dashboard and the accompanying analysis is structured around the comparison between the group stage performance under Jean-Louis Gasset and the Knockout Stage performance under Emerse Faé, which individual players drove the improvement, and underlying data suggesting whether the title was deserved. The data that has been used throughout this report was sourced from StatsBomb's open data repository, which has provided event-level data for the AFCON data. The dashboard is hosted publicly on GitHub Pages and built using R, ggplot2 and Plotly, with the full code available in the repository that has been linked.

3 Data Collection and Preparation

3.1 Data Source

All the data that has been used in the analysis was obtained from StatsBomb's open data repository, which was computed into R via the statsbombr package (STATSBOMB REFERENCE). This open data provided event level match data for AFCON, which allowed for extensive game by game analysis to be conducted. This was downloaded by: `devtools::install.github("statsbomb/StatsBombR")`.

The StatsBomb event data model captures every on the ball action in a match as its own event, each with a timestamp, location (expressed as an x,y coordinate on a 120 x 80 pitch), the player and the possession team involved, and many more action specific events. The three primary functions that were used to collect the data were **FreeCompetitions()**, **FreeMatches()** and **get.matchFree()**. **FreeCompetitions()** was used to retrieve the full list of available competitions and identified AFCON2023. To filter for this, the competition id was filtered using 1267. This allowed for the match data to be downloaded, which is where **FreeMatches()** was used. In order to pull event level data from the matches, **get.matchFree()** was used to return a database of every recorded event, such as shots, passes, dribbles and duels etc.

3.2 Dataset structure and coverage

The final merged dataset consisted of all 52 AFCON 2023 matches. The event level data frame consisted of 162,910 rows across all matches, with each row representing a single recorded event. The key variables used in this research were:

Variable	Type	Description
match_id	Integer	Unique match identifier
competition_stage.name	Character	Round of competition
team.name	Character	Team Name
type.name	Character	Event type
shot.statsbomb_xg	Double	Expected goals value
shot.outcome_name	Character	Shot result
player.name	Character	Full player name
location	List	x,y coordinates on the pitch
minute/second	Integer	Event timestamp
possession_team.name	Character	Team in possession
duel.type.name	Character	Type of duel occurring
pass.goal_assist	Logial	Whether the pass led to a goal

For Ivory Coast specifically, the **ivorycoast_matches** subset identified the seven matches that they played in. The events from these matches were filtered from **all_events** using the `match_id` to analyse specific matches. This allowed for specific event data to be pulled from their matches, and using `possession_team.name == "Côte d'Ivoire"` allowed for a more focused view on the team.

One important structural decision that was required to be made was the exclusion of `"period == 5"`, which corresponds to a penalty shootout in Statsbomb databases. This was done as the penalty shootout goals, shots and xG were being included into the overall shot and goal analysis. Including these events would have altered the results, as penalty shootouts occur outside of normal time, therefore they do not represent in game performance in the same way that the regular event data does. Penalties carry a very high xG value, which would mean that when analysing the cumulative xG for Ivory Coast throughout the tournament, the penalty shootout vs Senegal would have interfered with the data collection.

3.3 Cleaning and Preparation

3.4 Limitations of the Data

There are certain limitations of the data set which affected the research. Firstly, the Statsbomb open data does not include tracking data (player positions in between the events), which limits the amount of off the ball metrics. This includes the pressing intensity and the overall defensive structure. Linking to this, the xG model used will not take underlying metrics such as pressure on the attacker, or how close the defender is, meaning

4 Methodology

4.1 Going from raw data to dashboard

4.2 Metrics computed

4.3 Grouping

4.4 Dashboard Design

5 Findings

5.1 Tournament Overview and the Group Stage Struggle

Across the whole tournament, Ivory Coast played 7 matches, winning 4, drawing one (won the game via a penalty shootout) and losing twice. Their two defeats came in the group stage where they lost to Nigeria 1-0 and Equatorial Guinea 4-0 in the final group stage game, which triggered the managerial change. During the knockout stages, Ivory Coast won three games and drew one, which they won on penalties. They had performed at a much higher standard, and that will be analysed later in this section. Table x summarises Ivory Coast's full tournament timeline, showing who they had faced, what the score was and what the result of the game was.

	competition_stage.name	opponent	ivc_score	opp_score	result
1	Final	Nigeria	2	1	Win
2	Semi-finals	Congo DR	1	0	Win
3	Round of 16	Senegal	1	1	Draw
4	Quarter-finals	Mali	2	1	Win
5	Group Stage	Equatorial Guinea	0	4	Loss
6	Group Stage	Nigeria	0	1	Loss
7	Group Stage	Guinea-Bissau	2	0	Win

The underlying data from these matches tell us a different story. As shown in Figure 1, Ivory Coast had a steady accumulation of xG in the group stage. Across the entire group stage, they accumulated 3.39 xG per game, which comes to an average of 1.13 per game. Taking the results into consideration, we can see that they produced inconsistent performances throughout the group stage, with the two defeats. Despite losing 4-0 in the last game to Equatorial Guinea, the team had generated 2.06 xG, which is more than the first two games combined. The lack of xG within those first two games indicate a lack of attacking creativity and a difficulty in producing high quality opportunities. The majority of shots that were created accumulated a lower amount of xG, therefore indicating the likelihood of scoring was low. However in the

last game of the group stage, despite the result, they managed to create the chances but were not able to finish them off. This sharp defeat led to the managerial change, even though they narrowly progressed to the next stage.

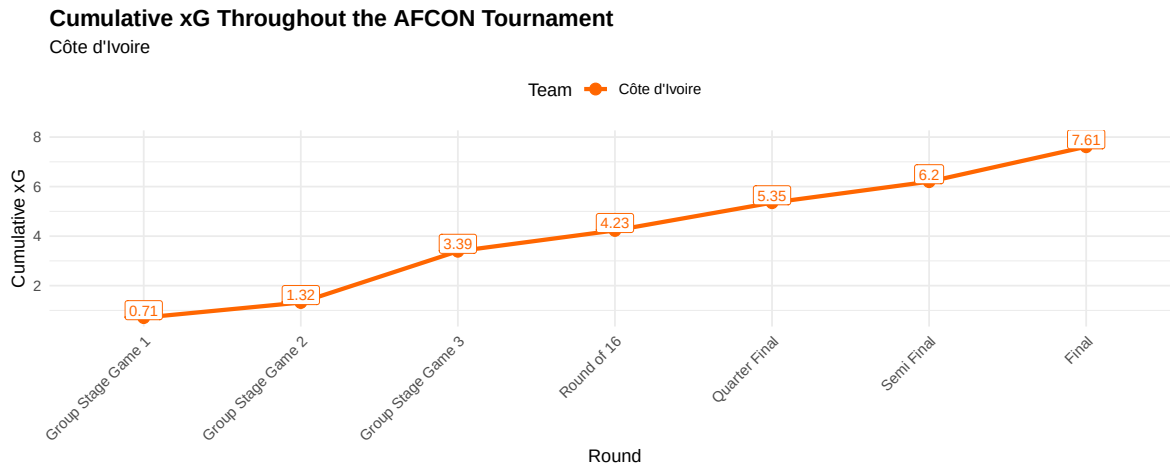


Figure 1: A graph highlighting the cumulative xG of Ivory Coast through each stage of the tournament

5.2 The Managerial Change: The introduction of Emerse Faé

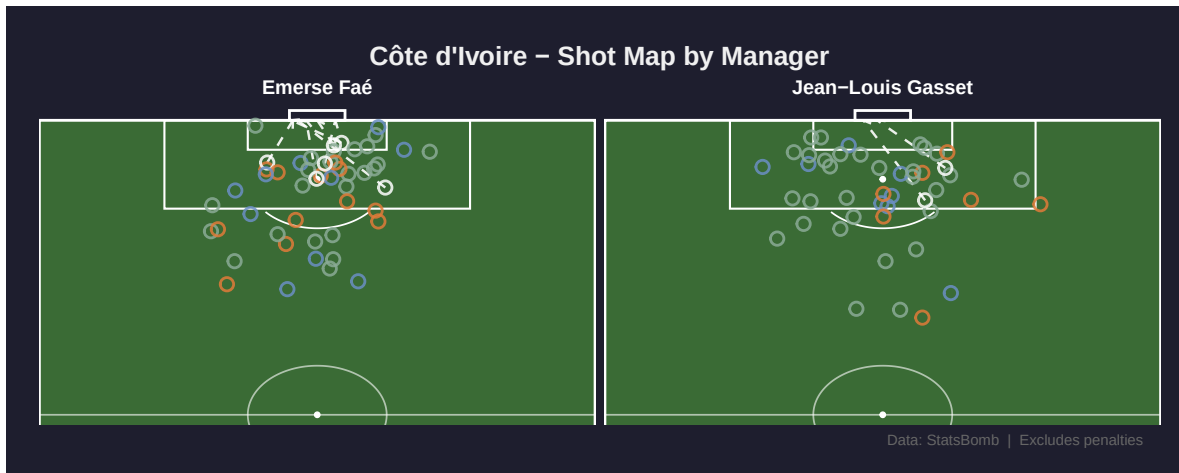
The appointment of Emerse Faé was a surprise for many reasons. No one expected a team to dismiss their manager during the season, especially as they had reached the knockout stages. However, the performances from the team under head coach Jean-Louis Gasset caused for immediate change. The managerial change produced measurable data evidenced changes across all metrics of Ivory Coast's performance. An examination of six areas of comparison between the two managers will best highlight the significant impact that the new manager created.

5.2.1 Shot Map and Attacking Locations

The shot maps in Figure 2 offer the clearest immediate illustration of how the team's attacking intent had changed.

Under Emerse Faé, Ivory Coast's shots appear to be more centrally concentrated inside and around the penalty area. a large proportion of these shots came in a dangerous area, which therefore gave them a higher chance of scoring. This suggests that there was a more direct attacking structure, which was focused on creating higher quality opportunities in central areas for their attackers. Additionally, the concentration of attacks near the goal indicates that they were frequently able to progress possession in advanced areas, whilst being able to finish the attacks (scoring or missing the chance).

In contrast, under Jean-Louis Gasset the shot distance is noticeably wider and more dispersed across the box. There are more attempts from wider angles, which the StatsBomb xG model assigns that the probability of those chances to be below 0.1. Goals were rare for the team under Gasset, and the two goals that they scored came from chances accumulating 0.075 xg and 0.056 xg respectively.



5.2.2 Shot Quality vs Quantity

The shot quality data shown in Table X formalises what the map shows in data form, illustrating a clear contrast between the two managers. Under Gasset, Ivory Coast averaged 15.3 shots per game at an average of 0.074 xG per shot. They had a shot on target percentage of 19.6%, with only 9 of their 46 shots being on target. Even though Jean-Louis Gasset's team averaged more shots per game, their output was far less effective. Faé's team scored 6 goals from their 51 shots, whereas Gasset's team only scored 2 from 46. These low amount of goals may not seem a lot, however, tournament football does not always provide high scoring games when teams of similar level play not to lose rather than win. There is a clear difference between avg xG per shot, where Fae's team averaged 0.083 per shot compared to Gasset's 0.074. On the surface level the difference may not seem that large, however over a large amount of shots it reflects a meaningful improvement in shot quality. This further supports the shotmap illustrating that the shots under Faé were in more dangerous areas.

A tibble: 8 x 3

Metric	`Emerse Faé`	`Jean-Louis Gasset`
<chr>	<chr>	<chr>
1 Matches	4	3
2 Total Shots	51	46
3 Shots on Target	16	9
4 Total xG	4.22	3.39

5 Goals	6	2
6 Shots per Game	12.8	15.3
7 Avg xG per Shot	0.083	0.074
8 Shot Accuracy (%)	31.4	19.6

The difference in shot on target percentage is huge, as Faé's team has an average of 31.4% and Gasset's has an average of 19.6%. This means that nearly one third of shots bt Faé's team tested the goalkeeper, while fewer than one in five did so under Gasset. An interesting contrast emerges between shot quantity and quality as Gasset's team attempted more shots but produced a lower shot on target percentage, lower average xG per shot, fewer goals and a lower total xG. This suggests that a higher shooting volume did not translate into higher attacking performance.

Overall, the table suggests that Ivory Coast were a more clinical and structually efficient attacking side under Faé. Although Gasset's team had generated more shots in less games, they were lower in xG, suggesting the shots were taken with more pressure and in positions that are less likely to test the goalkeeper. In contrast, Fae's team created higher quality chances, tested the goalkeeper more frequently and converted their chances more efficiently.

5.2.3 Shots Conceded and xG Against and Defensive Actions

The data suggests that there was a clear improvement in defensive perfomance following the managerial change. Under Gasset, Ivory Coast conceded more shots per game, and xG per shot with 10.3 shots per game and 1.07 xG per shot. This indicates that they had a vulnerable defensive structure due to the higher amount of high quality chances that they conceded.

In contrast, under Faé, there was a slight improvement. Even though they conceded more xG and more shots, the average for both were lower as their shots conceded per game was 9 and 0.93 xG per shot. This illustrates that they restructured their opponents to lower quality chances, relecting a more organised and resilient structure.

```
# A tibble: 5 x 3
```

Stat	`Jean-Louis Gasset`	`Emerse Faé`
<chr>	<chr>	<chr>
1 matches	3	4
2 shots_conceded	31	36
3 xg_conceded	3.2	3.73
4 shots_conceded_per_game	10.3	9
5 xg_conceded_per_game	1.07	0.93

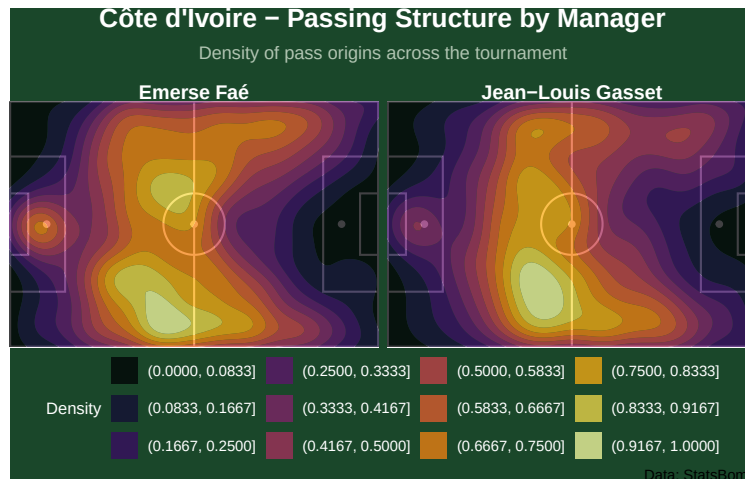
Furthermore, the defensive metrics used have reinforced the improvement in their structure. There has been a rise in tackles from 14.3 to 19.5, suggesting that they have gained more

success in tackles and are defending more as a unit. This can be linked to pressures which has also increased from 104 to 141.8 per game. This indicates that there has been a change in intensity, with a more aggressive approach which allows them to turnover possession high, and start a counter attack using their quick wingers. Duels per game have also increased from 26.7 to 36.5, which reflects a more physical approach in contests and ball winning situations in midfield and defense. Interceptions have increased from 6.3 to 7.8 per game which indicates that there has been an increased focus on closing down spaces and blocking passing lanes. Finally, clearances have increased from 11 to 26 per game which suggests that the defenders are prioritising clearing the ball away from dangerous situations rather than keep it in the attacking zones.

```
# A tibble: 6 x 3
  Stat      `Jean-Louis Gasset` `Emerse Faé`
  <chr>      <chr>          <chr>
1 matches      3              4
2 tackles_pg  14.3           19.5
3 pressures_pg 104            141.8
4 duels_pg     26.7           36.5
5 interceptions_pg 6.3           7.8
6 clearances_pg 11             26
```

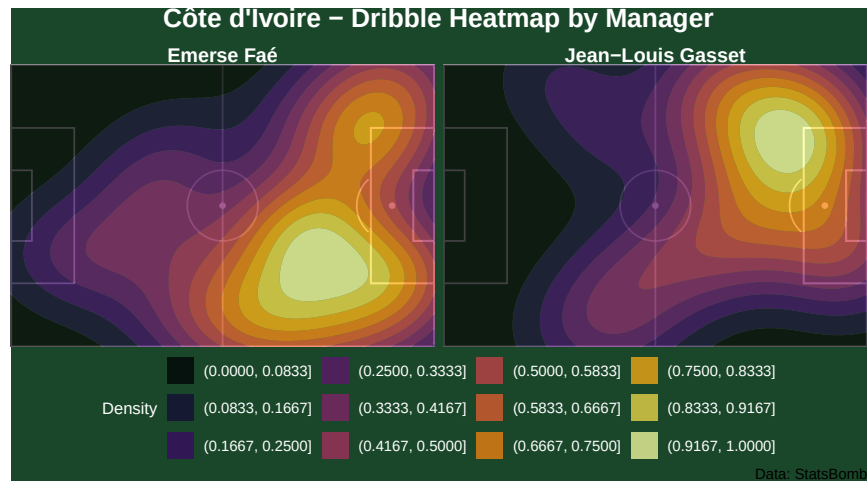
5.2.4 Passing Structure

Warning: `stat(level)` was deprecated in ggplot2 3.4.0.
i Please use `after\_stat(level)` instead.



5.2.5 Dribble Areas

Figure X compares Ivory Coast's dribble density under both managers.

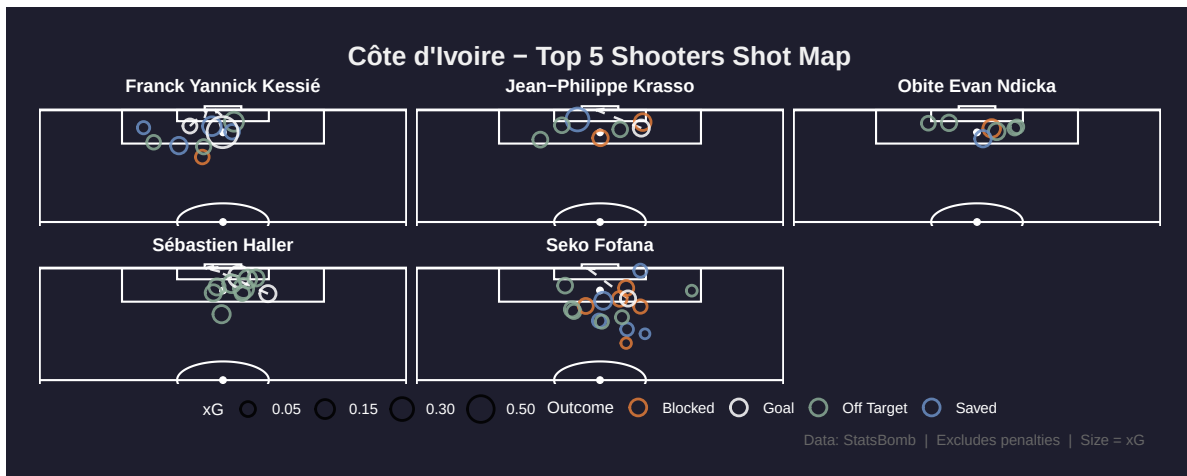


Under Faé, the dribble areas are heavily concentrated on the wide areas, with it being spread on both sides of the pitch. In heat map

5.3 Key player contributions

5.3.1 Shot Map

Figure X below illustrates the shot maps of the top five shooters for Ivory Coast.



As shown, Sébastien Haller's shots are heavily concentrated very close to the goals, which highlights his role as the main striker and the main focal point for all the attacks. In contrast,

we can see that the midfielders, Franck Kessie and Seko Fofana shower a much wider distribution of shots, as they are attempting shots from outside the box whilst also taking shots within the box. This indicates that they are making late runs into the box. Krasso also has shots in central areas as he was used as a backup striker to Haller. Evan Ndicka's limited amount of shots likely came from set piece situations.

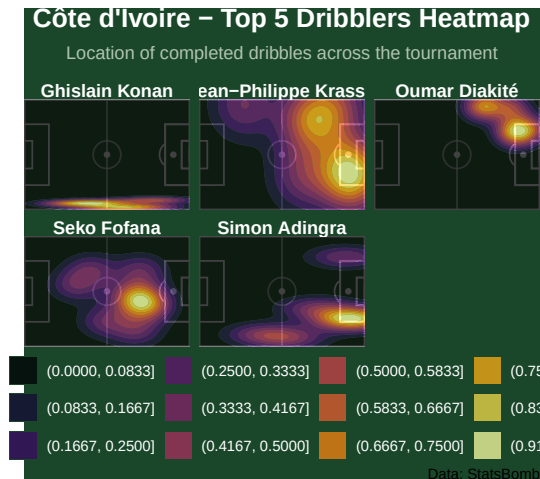
Overall, it appears from the shot maps that Ivory Coast had a well organised attacking strategy with distinct player responsibilities. A focus on effective opportunity generation and attacking via the middle of the pitch is indicated by the concentration of shots through dangerous central areas.

5.3.2 Goals and Assists

	player.name	Goals	Assists	GA
1	Franck Yannick Kessié	2	1	3
2	Sébastien Haller	2	0	NA
3	Jean-Philippe Krasso	1	0	NA
4	Oumar Diakité	1	0	NA
5	Seko Fofana	1	0	NA
6	Simon Adingra	1	2	3
7	Max-Alain Gradel	0	1	NA

From this goals and assists table, we can see that the goals have been spread the team, as no player has scored more than two goals. Franck Kessié and Simon Adingra recorded the highest amount of goal contributions with three each, which highlights their importance in both chance creation and goal scoring. Overall, the table shows that the team was very balanced in attack, and they did not have to rely on one singular player.

5.3.3 Dribbling Heat Maps



The dribbling heat maps show that Ivory Coast relied heavily on wide attacking dribbles. Oumar Diakité, Jean-Phillipe Krasso and Simon Adingra are the three players with the highest concentration in the wider areas. This further supports the point that they relied on wing play to attack the opponent and pin them back.

In contrast, we have Seko Fofana who plays in midfield meaning his dribbling concentration is more central. He used his dribbles to connect the defense and the attack, and progress the ball through midfield. Ghislain Konan showed deeper and wider dribbling on the left side of defense, which illustrated his importance in helping the team relieve pressure by dribbling out of tough situations.

5.3.4 Defensive stats

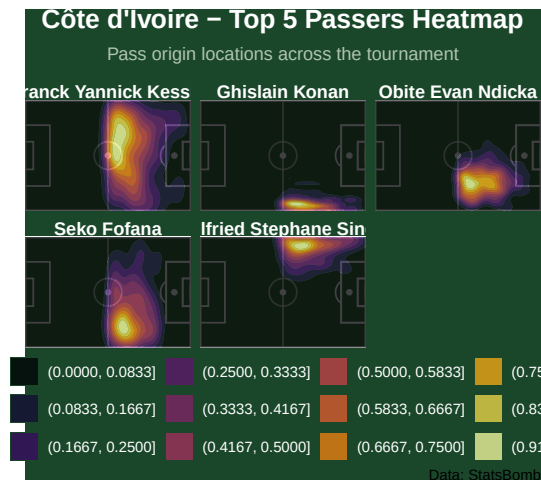
The defensive data highlights that Kessié and Seri were the most efficient defensive midfielders. Kessié recorded a very strong tackle success rate at 83.3% whilst Seri recorded 80%. These were both from a high number of tackles meaning that they were very efficient in the defensive roles that had been set for them. Sangaré also contributed massively, not only with tackles. All round he was a very solid defensive midfielder which allowed the attackers freedom to play.

In the wider defensive areas, both Konan and Singo were involved in a large number of duels and recoveries. Both recorded rather low percentages for their duels won, however, they both excelled at other areas such as interceptions and clearances. The low percentage duels won indicate they were facing difficult one on one situations.

Overall, the table highlights that their defensive strength came from the effectiveness of their midfield, which was supported by the aggressive nature of their fullbacks.

```
# A tibble: 22 x 12
  player.name      tackles tackles_won tackle_pct pressures interceptions
  <chr>          <int>      <int>      <dbl>    <int>          <int>
1 Ghislain Konan      19         8      42.1      66            8
2 Franck Yannick Kessié 12        10      83.3      89            4
3 Ibrahim Sangaré     10         7       70       96            4
4 Jean Michaël Seri    10         8       80       51            3
5 Wilfried Stephane Sin~ 9         3      33.3      36            7
6 Serge Aurier         8         5      62.5      34            6
7 Obite Evan Ndicka    7         6      85.7      48            1
8 Oumar Diakité       7         6      85.7      51            4
9 Willy Boly          7         1      14.3      28            1
10 Seko Fofana         6         3       50       63            1
# i 12 more rows
# i 6 more variables: inter_success <int>, inter_pct <dbl>, duels <int>,
#   duels_won <int>, duel_pct <dbl>, clearances <int>
```

5.3.5 Top 5 Passers



The passing heat maps indicate that Ivory Coast’s buildup play was heavily concentrated through central areas. Kessié and Fofana illustrated high passing activity in central areas, which highlights their importance in connecting the attack and defense with their progressive passes.

Distribution from the flanks was also important for Ivory Coast with Singo and Konan both having high concentration in wide areas. The teams use of fullbacks to support buildup allowed them to maintain width in attacks. Furthermore, Evan Ndicka had a huge role in progression as the high concentration suggests he had a key role in starting attacks.

5.3.6 Attacking Stats

This attacking data table highlights that Seko Fofana is one of the most influential offensive players. He ranks highly in four out of the 6 metrics, with only progressive passing and box entries being his week points. Simon Adingra and Oumar Diakaté both played major roles in attacks, with high amounts of in all metrics apart from progressive passes which indicates that they are always on the receiving end of progressive passes.

There is a huge reliance on the fullbacks in buildup with Konan, Aurier and Singo who have high entries into and touches in the final 3rd, with a huge amount of progressive passes. This further indicates that there was a huge reliance on width, and the players involved supported efficiently.

```
# A tibble: 24 x 7
  player.name      carries_final3rd box_entries dribbles touches_final3rd
  <chr>              <int>         <int>    <int>         <int>
1 Ghislain Konan      12             0        5          163
2 Seko Fofana         15             1        9          160
3 Oumar Diakité        7             3        6          146
4 Franck Yannick Kessié 10             1        4          135
5 Serge Aurier         6             0        1          133
6 Max-Alain Gradel     5             2        2          125
7 Simon Adingra        7             3        7          106
8 Nicolas Pépé         2             4        4          100
9 Wilfried Stephane Sin~ 6             1        0           91
10 Jeremie Boga         4             3        4           90
# i 14 more rows
# i 2 more variables: progressive_passes <int>, shot_assists <int>
```

5.4 Was the title deserved?

A metric was created to determine who deserved to win the tournament. A weighting was applied depending on what stage the team reached, with the final having a weight of 1, semi final 0.9, Quarter Final 0.85, Round of 16 0.8 and the Group Stage 0.75. This was done so that all factors could be taken into consideration, such as the amount of games they had to play and teams playing harder opponents later in the tournament. Attacking, progressive and defensive stats were also included in the metric.

```
# A tibble: 24 x 21
  team.name      competition_stage.name stage_weight matches goals xg_pg
  <chr>          <fct>              <dbl>    <int> <int> <dbl>
1 Côte d'Ivoire Final                1         7     8  1.09
```

2	Egypt	Round of 16	0.8	4	7	1.57
3	Nigeria	Final	1	7	7	1.39
4	Cape Verde Islands	Quarter-finals	0.85	5	8	1.36
5	South Africa	Semi-final → 3rd Place F~	0.9	7	7	1.07
6	Angola	Quarter-finals	0.85	5	9	1.33
7	Senegal	Round of 16	0.8	4	9	1.3
8	Congo DR	Semi-final → 3rd Place F~	0.9	7	6	1.63
9	Mali	Quarter-finals	0.85	5	5	1.17
10	Mauritania	Round of 16	0.8	4	3	0.93

```

# i 14 more rows
# i 15 more variables: shots_on_target_pg <dbl>, shot_assists_pg <dbl>,
#   progressive_passes_pg <dbl>, dribbles_pg <dbl>, carries_final3rd_pg <dbl>,
#   box_entries_pg <dbl>, touches_final3rd_pg <dbl>, tackles_won_pg <dbl>,
#   interceptions_pg <dbl>, clearances_pg <dbl>, pressures_pg <dbl>,
#   attack_score <dbl>, progressive_score <dbl>, defensive_score <dbl>,
#   overall_score <dbl>

```

The scores revealed that Ivory Coast did infact deserve to win the tournament despite their poor Group Stage performance as their underlying metrics improved significantly across the tournament. Their transition from the inconsistent group of performers to a more efficient team in the knockout stages is reflected in the high attacking score. All round, they are the most consistent team across the four key attributes.

Nigeria who also got to the final are in third place, suggesting that Ivory Coast's win in the final was deserved. While Nigeria were more consistent throughout the tournament, Ivory Coast's effective knockout stage, tied with the weighting of reaching the final, allowed for them to finish at the top of this chart.

Interestingly, Egypt ranked second in this chart despite finishing in the Round of 16. There are many factors to this as they were very strong in the group stages, however, key player Mohammed Salah suffered an injury meaning they were missing their strongest player for the knockout stages.

6 Discussion, Limitation and Conclusion

6.1 Implication of the Findings

6.2 Limitations

6.3 Future Outlook

6.4 Conclusion

7 References